Syed Hanzala Ali-24K0025-LAB05-Class tasks

Q1

[SQL Worksheet]*

```sql
CREATE TABLE employees AS SELECT * FROM hr.employees;
CREATE TABLE departments AS SELECT * FROM hr.departments;

SELECT COUNT(*) AS total_employees
FROM employees;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.081 seconds

	TOTAL_EMPLOYEES
1	107

```
FROM employees;

SELECT MAX(salary) AS max_salary
FROM employees;
```

Query result | Script output | DBMS output | Explain Plan

Download ▾ Execution time: 0.08 seconds

	MAX_SALARY
1	24000

```

SELECT MIN(salary)
FROM hr.employees;
```

Query result | Script output | DBMS ou

Download ▾ Execution tin

	MIN(SALARY)
1	2100

```
12
13  SELECT SUM(salary)
14  FROM hr.employees;
```

Query result | Script output | DBMS o

Download | Execution ti

	SUM(SALARY)
1	691416

```
14    FROM hr.employees;
15
16    SELECT department_id, COUNT(*)
17    FROM hr.employees
18    GROUP BY department_id;
```

Query result | Script output | DBMS output | Explain Plan

Download ▾ Execution time: 0.006 seconds

	DEPARTMENT_ID	COUNT(*)
1	90	3
2	60	5
3	100	6
4	30	6
5	50	45

Q2

```
SELECT department_id, AVG(salary)
FROM hr.employees
GROUP BY department_id;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	DEPARTMENT_ID	AVG(SALARY)
1	90	19333.333333333332
2	60	5760
3	100	8601.333333333334
4	30	4150
5	50	3475.5555555555557

```
SELECT department_id, COUNT(*)
FROM hr.employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

Query result | Script output | DBMS output | Explain Plan

Download ▾ Execution time: 0.005 seconds

	DEPARTMENT_ID	COUNT(*)
1	100	6
2	30	6
3	50	45
4	80	34

```sql
SELECT department_id, SUM(salary)
FROM hr.employees
GROUP BY department_id
ORDER BY SUM(salary) DESC;
```

Query result | Script output | DBMS output | Explain Plan | SQL histor

Download ▾ Execution time: 0.006 seconds

	DEPARTMENT_ID	SUM(SALARY)
1	80	304500
2	50	156400
3	90	58000
4	100	51608
5	60	28800

```sql
INSERT INTO departments (department_id, department_name, manager_id, location_id)
VALUES (300, 'Data Science', 100, 1700);

UPDATE employees
SET salary = salary * 1.05
WHERE department_id = 50;
```

Q3

```sql
SELECT department_id, SUM(salary), AVG(salary)
FROM hr.employees
GROUP BY department_id
HAVING AVG(salary) > 7000;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.001 seconds

	DEPARTMENT_ID	SUM(SALARY)	AVG(SALARY)
1	90	58000	19333.333333333332
2	100	51608	8601.333333333334
3	80	304500	8955.882352941177
4	20	19000	9500
5	70	10000	10000

```sql
DELETE FROM departments
WHERE department_id = 280;

SELECT department_id, COUNT(*)
FROM hr.employees
WHERE salary > 5000
GROUP BY department_id
HAVING COUNT(*) > 3;
```

Query result | Script output | DBMS output | Explain Plan | SQL hi

Download ▾ Execution time: 0.006 seconds

	DEPARTMENT_ID	COUNT(*)
1	100	6
2	50	5
3	80	34

```
INSERT INTO employees_backup
SELECT * FROM employees;

SELECT department_id, AVG(salary)
FROM hr.employees
WHERE job_id != 'IT_PROG'
GROUP BY department_id
HAVING AVG(salary) > 5000
ORDER BY AVG(salary) DESC;
```

Query result | Script output | DBMS output | Explain Pl

Download ▾ Execution time: 0.008 seconds

	DEPARTMENT_ID	AVG(SALARY)
1	90	333.333333333332
2	110	10154
3	70	10000
4	20	9500